

CYBER SECURITY PRACTICALS – COMPLETE VIVA + THEORY + COMMANDS GUIDE

Based on your experiment file

EXPERIMENT 1

DAC and RBAC in Linux

1. What We Need To Do In This Experiment

In this experiment, we learn how Linux controls access and permissions using:

1. DAC (Discretionary Access Control)
2. RBAC (Role-Based Access Control)

We create users and apply different permission methods.

Main Objective

- Learn file ownership
 - Learn Linux permissions
 - Learn role-based administration
 - Restrict unauthorized access
 - Understand security management in Linux
-

Real Life Example

System Real Example

DAC Your personal mobile gallery

RBAC Company employee ID access

Example:

- In DAC, Alice owns a folder and decides who can access it.
 - In RBAC, Bob gets permissions because of his role/job.
-

2. What We Are Using And About Them

Tool	Purpose
adduser	Create users
mkdir	Create directory
chown	Change ownership
chmod	Change permissions
groupadd	Create group
usermod	Add user into group
visudo	Edit sudo rules securely
sudo	Run commands as administrator

3. Definitions and Full Forms

Term	Full Form	Definition
DAC	Discretionary Access Control	Owner controls permissions
RBAC	Role-Based Access Control	Access based on job role
chmod	Change Mode	Changes file permissions
chown	Change Owner	Changes file owner
sudo	Super User Do	Run admin commands
Group	—	Collection of users
Permission	—	Access rights to files

Example

chmod 700 folder

Meaning:

- Owner → Full access
 - Others → No access
-

4. Proper Commands in Sequence

DAC Commands

```
sudo adduser alice
```

```
sudo adduser bob
```

```
sudo mkdir -p /home/alice/project
```

```
sudo chown alice:alice /home/alice/project
```

```
sudo chmod 700 /home/alice/project
```

```
su - alice -c "cd ~/project && touch alice_file.txt && ls"
```

```
su - bob -c "cd /home/alice/project"
```

RBAC Commands

```
sudo groupadd syshelper
```

```
sudo usermod -aG syshelper bob
```

```
sudo visudo
```

Add this line:

```
%syshelper ALL=(ALL) /usr/bin/apt
```

Test:

```
su - bob
```

```
sudo apt update
```

```
sudo useradd testuser
```

```
exit
```

Output Explanation

Command	Result
Alice access	Success
Bob access	Permission denied
sudo apt update	Allowed
sudo useradd	Denied

Difference Between DAC and RBAC

Feature	DAC	RBAC
Based On	Ownership	Role
Controlled By	File Owner	Administrator
Best For	Personal files	Organizations
Flexibility	High	Structured
Security	Moderate	Strong

5 Important Viva Questions

Q1. What is DAC?

DAC is a security model where file owners decide permissions.

Q2. What is RBAC?

RBAC gives permissions according to user roles.

Q3. Why use chmod 700?

It gives full permission only to owner.

Q4. What does sudo do?

It allows normal users to execute admin commands.

Q5. Why is RBAC more secure?

Because users only get permissions required for their job.

Important Teacher Questions

Question	Short Answer
What is Linux permission system?	Security access system
What is file ownership?	User control over file
What is group in Linux?	Collection of users
Why use visudo?	Safe sudo editing
Difference between user and role?	User is person, role is responsibility

EXPERIMENT 2

MAC using AppArmor

1. What We Need To Do

We use AppArmor to restrict access to files even when normal permissions allow access.

Objective:

- Learn MAC security
 - Learn AppArmor profiles
 - Understand Linux kernel security
-

Real Life Example

Suppose:

- A student has library access card
- But principal blocks entry into lab

Even though access exists, system restriction blocks it.

This is MAC.

2. What We Are Using

Tool	Purpose
AppArmor	Linux security framework
aa-status	Check AppArmor status
apparmor_parser	Load profiles
journalctl	Check logs
dmesg	Kernel messages
nano	Text editor

3. Definitions and Full Forms

Term	Full Form	Definition
MAC	Mandatory Access Control	System-controlled permissions
DAC	Discretionary Access Control	Owner-controlled permissions
AppArmor	Application Armor	Linux security module
Kernel	—	Core of operating system
Logs	—	Activity records

Why MAC is Stronger than DAC

DAC	MAC
User controls access	System controls access
Easier to bypass	Strong security
Based on permissions	Based on policies

4. Proper Commands in Sequence

Install AppArmor

```
sudo apt update
sudo apt install apparmor apparmor-utils -y
```

Start AppArmor

```
sudo systemctl enable apparmor
sudo systemctl start apparmor
sudo systemctl restart apparmor
```

Check Status

```
sudo aa-status
```

Create Folder

```
sudo mkdir /home/testfolder
```

Give Permissions

```
sudo chmod 777 /home/testfolder
```

Create Secret File

```
echo "secret information" | sudo tee /home/testfolder/secret.txt
```

Test Access

```
cat /home/testfolder/secret.txt
```

Create Profile

```
sudo nano /etc/apparmor.d/usr.bin.cat-block
```

Paste:

```
abi <abi/4.0>,  
include <tunables/global>  
profile /usr/bin/cat {  
    include <abstractions/base>  
    file,  
    deny /home/testfolder/** r,  
}
```

Load Profile

```
sudo apparmor_parser -r /etc/apparmor.d/usr.bin.cat-block
```

Verify Profile

```
sudo aa-status | grep cat
```

Test Restriction

cat /home/testfolder/secret.txt

Expected:

Permission denied

Check Logs

sudo journalctl | grep apparmor

OR

sudo dmesg | grep DENIED

5 Important Viva Questions

Q1. What is MAC?

Security model controlled by system policies.

Q2. What is AppArmor?

Linux security module that restricts programs.

Q3. Why did permission deny occur?

AppArmor profile blocked file access.

Q4. What is kernel?

Core part of operating system.

Q5. Difference between DAC and MAC?

DAC uses user permissions; MAC uses system policies.

EXPERIMENT 3

John The Ripper

1. What We Need To Do

We perform password cracking using John the Ripper.

Objectives:

- Learn password security
 - Understand password hashing
 - Test weak passwords
-

Real Life Example

Suppose:

- A company stores encrypted passwords
- Ethical hacker tests password strength

This is password auditing.

2. What We Are Using

Tool	Purpose
John The Ripper	Password cracker
md5sum	Create MD5 hash
hash.txt	Store hash
incremental mode	Brute-force attack

3. Definitions and Full Forms

Term	Full Form	Definition
MD5	Message Digest 5	Hashing algorithm

Term	Full Form	Definition
Hash	—	Encrypted fixed-length value
Brute Force	—	Try all combinations
Password Cracking	—	Recover original password

Proper Commands in Sequence

Create MD5 Hash

```
echo -n 'Ab' | md5sum
```

Save Hash

```
echo '0e4c46df226b9c0cb391311c54f28efe' > hash.txt
```

Start Cracking

```
john --format=raw-md5 --incremental hash.txt
```

Show Password

```
john --show hash.txt
```

Output

```
Ab
```

5 Important Viva Questions

Question	Answer
What is John The Ripper?	Password cracking tool
What is hash?	Encrypted password form
What is MD5?	Hashing algorithm

Question**Answer**

What is brute force?

Trying all combinations

Why use password cracking? Security testing

EXPERIMENT 4

MFA using Google Authenticator

1. What We Need To Do

We enable Multi-Factor Authentication in Linux SSH login using OTP.

Objectives:

- Secure SSH login
 - Learn PAM authentication
 - Use Google Authenticator
-

Real Life Example

ATM card:

- PIN
- OTP

Both required.

This is MFA.

2. What We Are Using

Tool	Purpose
Google Authenticator	Generate OTP
SSH	Remote login
PAM	Authentication module
OTP	One-time password

3. Definitions and Full Forms

Term Full Form

MFA Multi-Factor Authentication

OTP One-Time Password

SSH Secure Shell

PAM Pluggable Authentication Module

Proper Commands in Sequence

Update Packages

```
sudo apt update
```

Install Google Authenticator

```
sudo apt install libpam-google-authenticator -y
```

Setup MFA

```
google-authenticator
```

Answers:

y

y

y

n

y

Edit PAM File

```
sudo nano /etc/pam.d/sshd
```

Add:

```
auth required pam_google_authenticator.so
```

Edit SSH Config

```
sudo nano /etc/ssh/sshd_config
```

Ensure:

PasswordAuthentication yes

KbdInteractiveAuthentication yes

UsePAM yes

Restart SSH

```
sudo systemctl restart ssh
```

Check IP

```
ifconfig
```

Login

```
ssh kali@127.0.0.1
```

Login Process

1. Enter OTP
 2. Enter password
 3. Access granted
-

5 Important Viva Questions

Question	Answer
-----------------	---------------

What is MFA?	Multiple authentication factors
--------------	---------------------------------

What is OTP?	Temporary code
--------------	----------------

What is SSH?	Secure remote login
--------------	---------------------

Why use MFA?	Extra security
--------------	----------------

Question	Answer
----------	--------

What is PAM?	Authentication framework
--------------	--------------------------

EXPERIMENT 7

NAC using FreeRADIUS

1. What We Need To Do

We simulate Network Access Control using:

- FreeRADIUS
 - 802.1X
 - EAP authentication
 - Wireshark packet capture
-

Real Life Example

Office Wi-Fi:

- User enters credentials
 - RADIUS verifies
 - Access allowed
-

2. What We Are Using

Tool	Purpose
FreeRADIUS	Authentication server
Wireshark	Packet analyzer
radtest	Authentication testing
EAP	Authentication protocol

3. Definitions and Full Forms

Term	Full Form
NAC	Network Access Control

Term	Full Form
RADIUS	Remote Authentication Dial-In User Service
EAP	Extensible Authentication Protocol
EAP-MD5	MD5-based EAP
802.1X	Network authentication standard

Proper Commands in Sequence

Update Packages

```
sudo apt update
```

Install Tools

```
sudo apt install freeradius freeradius-utils wireshark -y
```

Configure User

```
sudo nano /etc/freeradius/3.0/users
```

Add:

```
student Cleartext-Password := "password123"
```

Stop Service

```
sudo systemctl stop freeradius
```

Start Debug Mode

```
sudo freeradius -X
```

Wait for:

Ready to process requests

Open Wireshark

sudo wireshark

Interface:

lo

Filter:

radius || eap

Authentication Test

```
radtest -t eap-md5 student password123 127.0.0.1 0 testing123
```

Expected Output

Access-Accept

Packets Captured

Packet

Access-Request

Access-Challenge

Access-Request

Access-Accept

5 Important Viva Questions

Question	Answer
What is NAC?	Network access security
What is RADIUS?	Authentication protocol
What is EAP?	Authentication framework
Why use Wireshark?	Capture packets
What is FreeRADIUS?	Open-source RADIUS server

MOST IMPORTANT FINAL VIVA QUESTIONS

Question	Answer
Difference between DAC and MAC?	DAC=user controlled, MAC=system controlled
What is RBAC?	Role-based permission system
What is AppArmor?	Linux security module
What is MFA?	Multi-layer authentication
What is RADIUS?	Authentication server protocol
What is Wireshark?	Packet analyzer
What is OTP?	Temporary authentication code
What is hash?	Encrypted data
What is brute force attack?	Trying all passwords
Why use cybersecurity practically?	To secure systems and networks

EXPERIMENT NO 8

Firewall Rule-Based Access Control using iptables

Aim

To learn how firewall rules work by:

- Blocking traffic
 - Allowing traffic
 - Logging traffic
 - Testing network connection using iptables in Linux.
-

1. What We Need To Do

In this experiment we:

1. Check connection between Windows and Linux VM
 2. Block ping requests
 3. View firewall logs
 4. Create temporary web server
 5. Block port 8000
 6. Allow port 8000 using rule order
 7. Remove all rules after experiment
-

Real Life Example

Firewall works like a security guard:

- Allows authorized users
- Blocks unauthorized users

Example:

- Office Wi-Fi allows employees only
 - Unknown users are blocked
-

2. What We Are Using

Tool	Use
iptables	Linux firewall
ping	Test connection
grep	Search logs
dmesg	View kernel logs
Python HTTP Server Temporary web server	

3. Definitions and Full Forms

Term	Full Form	Meaning
IP	Internet Protocol	Device address
ICMP	Internet Control Message Protocol	Used for ping
TCP	Transmission Control Protocol	Network communication
LOG	Logging	Store activity records
DROP	—	Block traffic
ACCEPT	—	Allow traffic
Firewall —		Network security system

4. Proper Commands in Sequence

```
# =====
```

```
# EXPERIMENT 8 - FIREWALL USING IPTABLES
```

```
# =====
```

```
# ----- Phase 0 : Check IP -----
```

```
ip a
```

```
# Note your Linux IP address
```

```
# Example:
```

```
# 10.0.2.15
```

```
# ----- Windows CMD -----
```

```
# Run this from Windows Host
```

```
ping 10.0.2.15
```

```
# =====
```

```
# Phase 1 : Block Ping + Enable Logging
```

```
# =====
```

```
sudo iptables -A INPUT -p icmp --icmp-type echo-request -j LOG --log-prefix  
"FW_PING_DROP: "
```

```
sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```

```
# ----- Test Ping Again from Windows CMD -----
```

```
ping 10.0.2.15
```

```
# =====
```

```
# Phase 2 : View Firewall Logs
```

```
# =====
```

```
sudo journalctl -k | grep "FW_PING_DROP"
```

```
# OR
```

```
sudo dmesg | grep "FW_PING_DROP"
```

```
# =====
```

```
# Phase 3 : Create Temporary Web Server
```

```
# =====
```

```
python3 -m http.server 8000
```

```
# ----- Open Browser on Windows -----
```

```
# Open this in browser:
```

```
# http://10.0.2.15:8000
```

```
# ----- Stop Server -----
```

```
# Press:
```

```
Ctrl + C
```

```
# =====
```

```
# Block Port 8000
```



```
# =====
```

```
sudo iptables -A INPUT -p tcp --dport 8000 -j DROP
```

```
# ----- Start Server Again -----
```

```
python3 -m http.server 8000
```

```
# ----- Refresh Browser -----
```

```
# Website should NOT open
```

```
# ----- Stop Server -----
```

```
# Press:
```

```
Ctrl + C
```

```
# =====
```

```
# Allow Port Using Rule Precedence
```

```
# =====
```

```
sudo iptables -I INPUT 1 -p tcp --dport 8000 -j ACCEPT
```

```
# ----- Check Rule Order -----
```

```
sudo iptables -L --line-numbers
```

```
# =====
```

```
# Start Server Again
```

```
# =====
```

```
python3 -m http.server 8000
```

```
# ----- Refresh Browser -----
```

```
# Website should open now
```

```
# ----- Stop Server -----
```

```
# Press:
```

```
Ctrl + C
```

```
# =====
```

Phase 4 : Cleanup

=====

sudo iptables -F

----- Verify Cleanup -----

sudo iptables -L

Difference Between ACCEPT and DROP

ACCEPT	DROP
Allows traffic	Blocks traffic
Connection works Connection fails	

Difference Between -A and -I

-A	-I
Add at bottom	Add at top
Lower priority Higher priority	

5 Important Viva Questions

Question	Answer
What is firewall?	Security system for network traffic
What is iptables?	Linux firewall tool
What is ICMP?	Protocol used for ping

Question	Answer
Why ping failed?	Firewall blocked packets
Why website worked again? ACCEPT rule came first	
